

# Java Program: IllegalArgumentException Handling

## 1. Aim

To write a Java program that demonstrates an IllegalArgumentException by passing an invalid argument to a method and handles the exception using a try-catch block.

## 2. Description

An IllegalArgumentException occurs when a method receives an argument whose value is not valid for the operation. In this program, the setAge() method accepts an age as an argument. If the age is negative, the method explicitly throws an IllegalArgumentException with the message "Age cannot be negative." The main method passes -5, causing the exception. The catch block catches the exception and displays its message.

## 3. Algorithm

1. Start the program.
2. Define the setAge() method with an integer parameter age.
3. Check whether age is less than zero.
4. If age is negative, throw an IllegalArgumentException.
5. In the main method, call setAge() with the value -5.
6. Catch the IllegalArgumentException using a catch block.
7. Display the exception message.
8. Stop the program.

## 4. UML Diagram

| IllegalArgumentException     |
|------------------------------|
| + setAge(age : int) : void   |
| + main(String[] args) : void |

## 5. Source Code with Comments

```
public class IllegalArgumentException {  
    // Method to validate the age  
    public static void setAge(int age) {  
        // Check whether the age is negative  
        if (age < 0) {  
            // Throw an exception for an invalid argument  
            throw new IllegalArgumentException("Age cannot be negative.");  
        }  
    }  
  
    public static void main(String[] args) {  
        try {  
            // Pass an invalid negative age  
            setAge(-5);  
        } catch (IllegalArgumentException e) {  
            // Handle the invalid argument exception  
            System.err.println(  
                "Caught IllegalArgumentException: " + e.getMessage()  
            );  
        }  
    }  
}
```

## 6. Compilation and Execution Commands

```
javac IllegalArgument.java  
java IllegalArgument
```

The first command compiles the Java source file. The second command executes the compiled program.

## 7. Output Screenshot

### **IllegalArgument - Console Output**

```
Caught IllegalArgumentException: Age cannot be negative.
```

## 8. Result

The Java program was successfully executed. It demonstrated an `IllegalArgumentException` when a negative age was passed to the `setAge()` method, and the exception was successfully handled using a try-catch block.